

Slide 1 — Building an Early Warning System for Financial Stress

In this slide, we summarize the problem.

We want to identify weeks of financial stress using anomaly detection techniques. The idea is to distinguish between normal periods, meaning stable or risk-on market conditions, and anomalous periods, which represent stress or risk-off conditions.

The dataset contains 1111 weekly observations, from 2000 to 2021, and 42 market variables: equity indices, bond indices, interest rates, currencies, commodities, volatility, and macro-financial indicators. The target variable identifies normal and anomalous weeks. We have 237 anomalous weeks, equal to 21.33% of the sample.

The objective is not to predict crises in the strict sense, but to recognize the emergence of stress conditions using the information available at that moment.

Slide 2 — Labelled Anomalies Capture Market Stress Episodes

In this slide, we show the MSCI USA index over time, with anomalous weeks highlighted.

The graph serves as a visual check of the quality of the target variable: the vertical bands indicate the weeks in which the target variable is equal to 1.

The anomalies are not randomly distributed, but concentrate around difficult market phases, characterized by drawdowns or higher instability.

This confirms that the target does not simply represent statistically unusual observations, but periods consistent with financial stress conditions.

Slide 3 — Stationarity Analysis Before Modelling

Before moving to the models, we need to check whether the financial variables can be used directly or whether they need to be transformed.

This is important because many anomaly detection models try to learn a stable distribution of normal observations. If we use non-stationary variables in levels, the model could confuse a long-term trend with a true anomaly.

In the graph, we see the example of MSCI USA: in levels, it shows a clear growth trend, while the differences are more stable around a constant mean, although outliers and heavy tails remain, as is typical of financial data.

We then applied the Augmented Dickey-Fuller test to all 42 variables. The result is clear: only 3 variables out of 42 are stationary in levels, while after the first difference all 42 variables become stationary.

This confirms that feature transformation is necessary to prevent the models from learning trends instead of stress signals.

Slide 4 — Transforming Market Variables into Stable Features

After the stationarity analysis, we chose the most appropriate transformation for each type of variable, taking into account both statistical properties and financial meaning.

For equity indices, bond indices, currencies, and positive commodities, we use log-returns, because these variables are positive and relative changes are more informative than absolute levels.

For interest rates and bond yields, instead, we use first differences, because they can be close to zero or negative, so log-returns are not always applicable.

The graph of the German 2-year Bund shows this clearly: there are 366 weeks with negative yield, equal to 32.9% of the observations. This justifies the use of first differences for this category.

Finally, VIX and macro surprise indicators are kept in levels, because their value is already directly informative about stress or macro-financial surprise conditions.

At this point, the dataset is ready for the following models: the variables are more stable, comparable, and financially coherent.

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In this first step, we build the baseline anomaly detector using a **Multivariate Gaussian model**. The idea is to estimate what “normal market behaviour” looks like, so the model is trained only on normal observations. We use **80% of normal observations for training**, while the cross-validation set contains **10% of normal observations and 50% of anomalies**, and the test set contains the remaining **10% of normal observations and 50% of anomalies**. This split allows the model to learn normality, while validation and test sets are used to tune and evaluate anomaly detection. After the split, we standardize the variables, estimate the Gaussian mean and covariance matrix, and check covariance stability through the **condition** number. The value is around **776**, so the covariance matrix is reasonably stable. Each validation observation receives a **log-probability score**: lower values indicate observations less compatible with normal market behaviour. The threshold is selected by maximizing **F2**, keeping precision above 0.7. **F2 gives more weight to recall, which fits early warning: it is better to accept some false alarms than to miss true anomalies.** The selected threshold achieves **F2 = 0.832**, with precision **0.723** and recall **0.864** on the cross-validation set.

Here we evaluate the MVG baseline on the test set, using the validation threshold. The key result is **F2 = 0.8519**, confirming that the model is well aligned with the early-warning objective. The model detects **107 out of 119 anomalies**, missing only **12 anomalous periods**, with recall around **0.90**. The trade-off is that it produces **45 false positives**, so the detector is sensitive but not perfectly selective. The **ROC-AUC is 0.771**, showing a reasonable ability to rank anomalous observations above normal ones. Overall, the MVG baseline is a strong starting point: it captures most anomalies and achieves a high F2 score.

First, we use Ledoit-Wolf shrinkage, a regularization method that makes the covariance matrix more stable and less noisy. Ledoit-Wolf reduces the condition number from **776 to 188**, making the covariance matrix about **4 times more stable**. However, classification performance is almost unchanged: recall remains **0.8992**, while F2 slightly decreases from **0.8519 to 0.8479**. So Ledoit-Wolf improves numerical stability, but not predictive performance. Then we test **RobustScaler**, which uses median and interquartile range instead of mean and standard deviation, reducing the influence of extreme values. Results are again very similar: precision slightly improves from **0.6903 to 0.6974**, but recall decreases from **0.8992 to 0.8908**, and F2 slightly decreases to **0.8439**. Overall, these checks confirm that the MVG baseline is already reasonably stable, and **StandardScaler + Ledoit-Wolf** remains the most balanced version.

The domain-informed MVG adds financial structure to the model. We split variables into **six asset-class groups**, fit one MVG per group, and combine the group anomaly scores. Then we add a VIX filter, so the model only flags anomalies during high-volatility periods. This makes the model more selective and easier to interpret financially. Precision rises to **0.8353**, and false positives fall to **14**. However, this selectivity has a cost: recall drops to **0.5966**, because the model misses **48 true anomalies**. Since F2 gives more weight to recall, F2 decreases strongly to **0.6328**. So the domain-informed model is precise and conservative, but not sensitive enough for the main early-warning objective.

Ledoit-Wolf is the same MVG model, but with a more stable covariance matrix.

The Gaussian model needs the covariance matrix to compute anomaly scores. If this matrix is unstable, the scores can be unreliable. Ledoit-Wolf regularizes it slightly. In our case, it improves stability, but the performance is almost unchanged, so it is mainly a robustness check.

Domain-informed MVG makes the model more financial and selective.

Instead of using all variables together, it groups them by asset class and computes anomaly scores inside each group. Then it adds a VIX filter, so an alarm is triggered only when the anomaly score is high and market volatility is high. This increases precision, but reduces recall, so it misses too many anomalies for an early-warning system.

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Slide 11 — Feature contribution to anomaly score

Now we move from pure model performance to interpretation. Up to this point, the MVG model gives us an anomaly score and a binary decision: normal or anomalous. However, for an Early Warning System, this is not enough. A risk manager does not only need to know that an alarm is raised; they also need to understand what is driving that alarm.

For this reason, we decomposed the Mahalanobis-distance-based anomaly score feature by feature. In simple terms, the model measures how far a weekly observation is from normal market behaviour, and this decomposition tells us which variables are responsible for that distance.

The bar chart shows the average contribution of each variable across the anomalies detected in the test set. The higher the bar, the more that variable pushes the observation away from the normal regime. We can interpret this as the MVG equivalent of a feature-importance or SHAP-style explanation.

Financially, the result is coherent with risk-off theory. VIX is one of the strongest contributors, which makes sense because implied volatility typically spikes during market stress. Equity-related variables, rates, bond-market variables and commodities also appear among the important drivers. This confirms that the model is not detecting random statistical noise, but market configurations that are economically meaningful.

Slide 12 — What crises did we detect?

After interpreting which features drive the anomaly score, the next question is: what kind of events are we actually detecting?

Here, the objective is to connect the statistical output of the model with real financial history. We therefore map the anomalous weeks onto the price history of the MSCI USA index and compare them with known crisis episodes. The main idea is that, if our Early Warning System is meaningful, the anomalies should cluster around periods of actual financial stress.

In the notebook, we explicitly considered well-known episodes such as the dot-com bust, the Global Financial Crisis, the European sovereign debt crisis, the China slowdown and the Covid-19 crash. The model detects many anomalies around these periods, which is an important validation check: it means that the system is capturing relevant market stress regimes.

However, not all anomalous weeks fall exactly inside the annotated crisis windows. This does not necessarily mean that they are false positives. Some of them may correspond to smaller stress episodes, temporary market corrections, or early signals that did not become full crises. This is important from a financial perspective because an Early Warning System should detect stress at its beginning, not only after the crisis is already obvious.

We also applied K-means clustering to the anomalous weeks. The purpose was to check whether all crises look the same or whether there are different types of abnormal regimes. The answer is that anomalies are not homogeneous. The clustering suggests that risk-off episodes can have different structures: for example, acute panic episodes, more prolonged market stress, or regimes driven by specific asset classes.

Slide 13 — Crisis timeline and cluster visualization

This slide visualizes the previous idea.

On the left-hand chart, the blue line represents the MSCI USA price index over time. The vertical shaded areas represent weeks classified as anomalous, while the annotated regions correspond to known historical crises. The key point is that many detected anomalies overlap with major drawdown periods. This gives financial credibility to the model because the alarms are concentrated around moments when markets were actually under pressure.

On the right-hand chart, the anomalies are grouped into different clusters. The colors represent different types of anomalous weeks identified by K-means. This is useful because a binary signal, normal versus anomaly, is too limited for practical risk management. A risk officer would benefit not only from knowing that markets are abnormal, but also from understanding what kind of abnormality is occurring.

For example, an equity-driven crash may require a different portfolio response compared with a rates-driven shock or a volatility-driven panic. So this slide shows that the Early Warning System can be extended from simple detection to crisis characterization.

Slide 14 — PCA and UMAP

In this slide, we analyse the geometry of the classification problem.

First, we used PCA to project the original high-dimensional feature space into two principal components. The objective was not to build the model using only two dimensions, but to visualize whether normal and anomalous observations are clearly separated. In the PCA plot, points are colored according to the classification outcome: true positives, false positives, false negatives and true negatives.

The important result is that anomalies do not simply sit at the extreme borders of the PCA space. Instead, they are mixed with normal observations. This means that abnormality is not just a matter of being far away along the main directions of variance. Some anomalies may be hidden in regions that look dense in a two-dimensional projection.

Then we used UMAP, which is a nonlinear dimensionality reduction method. Unlike PCA, UMAP can capture more complex manifold structures. However, the conclusion is similar: anomalies are still interspersed with normal points. This suggests that the normal-versus-abnormal distinction is not linearly separable and cannot be fully understood through a simple two-dimensional view.

This is an important modelling insight. It explains why more flexible models may be useful later: the boundary between normal and abnormal market regimes is complex, nonlinear, and driven by interactions across several financial variables.

Slide 15 — Comparing models

Finally, before moving to the next modelling section, we define how the models should be compared.

Since this is an Early Warning System, the choice of evaluation metrics is not neutral. Precision tells us how reliable the alarms are: among the weeks flagged as anomalous, how many are truly anomalous? Recall tells us how many actual stress periods we detect. In this business case, recall is especially important because missing a crisis is more costly than raising a false alarm.

For this reason, we use the F2 score as the main optimization target. Compared with the F1 score, F2 gives more weight to recall. This reflects the business asymmetry of the problem: in portfolio risk management, a false negative can expose the investor to large losses, while a false positive may only lead to a temporary defensive allocation.

We also use ROC-AUC to evaluate the ranking quality independently of a specific threshold, and the confusion matrix to see the exact breakdown between true positives, false positives, false negatives and true negatives.

So the key message of this section is that model evaluation must be aligned with the business objective. We are not simply looking for the highest generic accuracy; we are looking for a model that detects financial stress early, keeps false alarms under control, and remains interpretable enough to be useful in a real risk-management process.